Effect of obstructive sleep apnoea and its treatment with continuous positive airway pressure on the prevalence of cardiovascular events in patients with acute coronary syndrome (ISAACC study): a randomised controlled trial

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Summary

Background Despite the improvement in the prognosis of acute coronary syndrome (ACS), substantial morbidity and mortality remain. We aimed to evaluate the effect of obstructive sleep apnoea (OSA) and its treatment with continuous positive airway pressure (CPAP) on the clinical evolution of patients with ACS.

Methods We designed a multicentre, open-label, parallel-group, randomised controlled trial of patients with ACS at 15 hospitals in Spain. Eligible non-sleepy patients were men and women aged 18 years and older, admitted to hospital for documented symptoms of ACS. All patients underwent respiratory polygraphy during the first 24–72 h after admission. OSA patients were randomly assigned (1:1) to CPAP treatment plus usual care (CPAP group) or usual care alone (UC group) by a computerised system available 24 h a day. A group of patients with ACS but without OSA was also included as a reference group. Because of the nature of the intervention, the trial intervention could not be masked to either investigators or patients. Patients were monitored and followed for a minimum of 1 year. Patients were examined at the time of inclusion; after 1 month, 3 months, 6 months, 12 months, 18 months, 24 months, 30 months, and 36 months; and every 12 months thereafter, if applicable, during the follow-up period. The primary endpoint was the prevalence of a composite of cardiovascular events (cardiovascular death or non-fatal events [Acute myocardial infarction, non-fatal stroke, hospital admission for heart failure, and new hospitalisations for unstable angina or transient ischaemic attack]) in patients followed up for a minimum of 1 year. The primary analysis was done according to the intention-to-treat principle. This study is registered with Clinicaltrials.gov, NCT01335087 and is now closed.

Findings Between April 25, 2011, and Feb 2, 2018, a total of 2834 patients with ACS had respiratory polygraphy, of whom 2551 (90·01%) were recruited. 1264 (49·55%) patients had OSA and were randomly assigned to the CPAP group (n=633) or the UC group (n=631). 1287 (50·45%) patients did not have OSA, of whom 603 (46·85%) were randomly assigned to the reference group. Patients were followed up for a median of 3·35 years (IQR 1·50–5·31). The prevalence of cardiovascular events was similar in the CPAP and UC groups (98 events [16%] *vs* 108 events [17%]; hazard ratio [HR] 0·89 [95% CI 0·68–1·17]; p=0·40) during follow-up. Mean time of adherence to CPAP treatment was 2·78 h/night (SD 2·73). The prevalence of cardiovascular events was similar between patients in the reference group (90 [15%] events) and those in the UC group (102 (17%) events) during follow-up (1·01 [0·76–1·35]; p=0·93). The prevalence of cardiovascular events seem not to be related to CPAP compliance or OSA severity. 464 (74%) of 629 patients in the CPAP group had 1538 serious adverse events and 406 (65%) of 626 patients in the UC group had 1764 serious adverse events.

Interpretation Among non-sleepy patients with ACS, the presence of OSA was not associated with an increased prevalence of cardiovascular events and treatment with CPAP did not significantly reduce this prevalence.

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Introduction

Acute coronary syndrome (ACS) is a major health burden, with more than 1 million patients suffering from ACS annually in the USA.[1] The short-term and long-term prognoses following diagnosis with ACS have improved (with acute revascularisation and treatment of known cardiovascular risk factors). However, substantial morbidity and mortality remain, including repeat cardiovascular events.

Obstructive sleep apnoea (OSA) is a common disease that affects 20–30% of the adult population[2] and is caused by the collapse of the upper airway during sleep. OSA has

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See **Online** for appendix

Research in context

Evidence before this study

Obstructive sleep apnoea (OSA) is a common condition among patients with cardiovascular disease, affecting 40–60% of such patients. Because the risk of recurrent cardiovascular events among these patients remains high despite contemporary therapies, CPAP could be a useful additional treatment for the prevention of these events. We searched PubMed with the terms "obstructive sleep apnea", "acute coronary syndrome", "CPAP treatment", and "secondary prevention" for systemic reviews and randomised controlled trials published up to Jan 1, 2019. Previous randomised controlled trials have shown that the treatment of OSA with CPAP is associated with a reduction in blood pressure, left ventricular mechanical overload, and arrhythmia. In addition, observational clinical studies have shown that the use of CPAP is associated with lower prevalence of cardiovascular complications and of death from cardiovascular causes. Three previous randomised controlled trials have directly addressed cardiovascular clinical endpoints in patients who have suffered a major cardiovascular event and the authors observed a neutral effect of CPAP treatment for secondary cardiovascular prevention.

Added value of this study

Our findings indicate that the use of CPAP therapy, compared with usual care alone, had no positive effect on the prevention of recurrent major cardiovascular events in patients with OSA and acute coronary syndrome. Moreover, the presence of OSA was not associated with an increased prevalence of cardiovascular events.

Implications of all the available evidence

The results of this secondary prevention trial in adults with acute coronary syndrome and OSA, supported by further studies, contribute to understanding the observed neutral effect of CPAP treatment for secondary cardiovascular prevention.

been associated with an increase in oxidative stress, inflammation, hypercoagulability, and sympathetic activation that could induce cardiovascular disease.[3–4] Epidemiological studies have shown that a high proportion (40–60%) of patients with coronary artery disease have OSA.[5–7] Previous randomised controlled trials have shown that the treatment of OSA with continuous positive airway pressure (CPAP) is associated with a reduction in blood pressure,[8] left ventricular mechanical overload, and arrhythmia.[9–11] These effects could improve the prognosis of patients with coronary artery disease.[12] However, other studies have shown that the use of CPAP was not associated with reduced risk of cardiovascular outcomes or death for patients with sleep apnoea.[13] Evidence suggests that in patients with chronic cardiovascular disease,[14] CPAP treatment is not effective as secondary cardiovascular prevention. To gain insight in this area of research,[15–18] we designed a secondary prevention trial to evaluate the effect of OSA and the effectiveness of its treatment with CPAP on the clinical evolution and prognosis of patients with ACS.[19]

Methods

Study design and patients

The ISAACC study (Impact of Sleep Apnea syndrome in the evolution of Acute Coronary syndrome. Effect of intervention with CPAP) is a multicentre, open-label, parallel-group, randomised controlled trial of patients with ACS at 15 hospitals across Spain (appendix p 5) The detailed design of this study has been published previously.[19]

Eligible patients were aged ≥18 years, had been admitted to hospital for ACS with a hospital stay of 24–72 h at the time of sleep study, and had an Epworth Sleepiness Scale (EES) score of ≤10 (ie, patients without excessive daytime sleepiness or non-sleepy). The exclusion criteria included previous treatment with CPAP for OSA; psychophysical inability to complete questionnaires; presence of any previously diagnosed sleep disorder; more than 50% central apnoeas or the presence of Cheyne–Stokes respiration; daytime sleepiness (ESS >10); chronic diseases; a medical history that could interfere with the study objectives or compromise conclusions; any medical factor, social or geographical, that might jeopardise patient compliance; any processes that reduce life expectancy to less than 1 year; and cardiogenic shock. Full eligibility criteria and definitions and the detailed reasons for exclusion during screening are provided in the appendix (p 5).

The Hospital Universitario Arnau de Vilanova in Lleida was the coordinator centre and was responsible for the overall management of the trial. The protocol was approved by the relevant regulatory authorities and ethics committees at each participating site, and patients provided written informed consent.

Randomisation and masking

Patients with OSA were randomly assigned (1:1) to receive either CPAP therapy plus usual care (CPAP group) or usual care alone (UC group). Simple randomisation was done by a computerised system available 24 h a day via a fully secure, password-protected study website. It was not possible to modify the result in any case. Each time we registered a patient in the database, we randomly assigned them by clicking on the button of the data collection web form. This click produced a call to a command or function "rand ()" within the source code with which the application was implemented, returning a random value. This function was part of the mathematics function library of the standard programming language PHP version 7.3. All used the random number generator of the processor in

which the call was executed. That is, rand (minInteger, maxInteger) returned an integer random value between a minInteger and maxInteger, depending on the number of groups of the randomisation. In this case, rand (0,1) returned 0 or 1. As the frequency of patients with AHI less than 15/h is higher than that of patients with AHI at least 15/h and the rate of inclusion into the reference group must match that of the randomised patients, a probabilistic mechanism integrated into the website selected patients once the sleep-test results were known. Because of the nature of the intervention, the trial intervention could not be masked to either investigators or patients.

Procedures

Eligible patients had respiratory polygraphy (Embletta, ResMed, Bella Vista, NSW, Australia) during the first 24–72 h after admission to hospital to diagnose OSA according to the national guidelines.[20] Apnoea was defined as an absence of airflow lasting 10 s or more. Hypopnoea was defined as a reduction in airflow lasting 10 s or more, associated with oxygen desaturation. Oxygen desaturation was defined as a decrease in arterial oxygen saturation of more than 4%. Respiratory polygraphy studies were done without supplemental oxygen. The AHI was defined as the number of episodes of apnoea and hypopnoea per h of recording. The degree of self-reported sleepiness was analysed by the validated Spanish version of the EES.[20]

Patients diagnosed with OSA (AHI ≥15/h) were then randomly assigned to the CPAP group or UC group. Patients without OSA (AHI <15/h) were randomly selected to be included in the reference group by means of probabilistic mechanism integrated into the web selected patients once the sleep-test results were known. CPAP pressure was titrated by a trained nurse using automated equipment before discharge, following a validated methodology.[21] Patients were instructed in hygienic-dietary measures and in the management of cardiovascular risk factors associated with lifestyle (details are specified in the appendix p 7). All patients were monitored and followed up for a minimum of 1 year. Clinic visits were scheduled for all patients at 1, 3, 6, 12, 18, 24, 30, and 36 months, and annually thereafter.

Sociodemographic and anthropometric variables were recorded, along with those associated with quality of life (EuroQol [EQ-5D]an instrument for measuring quality of life) and degree of daytime sleepiness. At baseline and after a 1-year follow-up, blood samples were obtained during fasting. CPAP adherence was measured by means of the machines' internal clocks. The degree of compliance was determined by dividing the number of hours of use by the number of days of treatment. Good compliance was defined as CPAP use of at least 4 h a day. Additional details on study measurements and monitoring procedures are provided in the appendix (pp 6–7, 12). Additional details on endpoint adjudication and definitions of primary outcomes are provided in the appendix (p 9-12).

Outcomes

The primary endpoint of the study was the prevalence of a composite of the first cardiovascular events (cardiovascular death or non-fatal events [acute myocardial infarction, non-fatal stroke, hospital admission for heart failure, and new hospitalisations for unstable angina or transient ischaemic attack]). A committee (see appendix p 11), whose members were unaware of the study group assignments, adjudicated the major cardiovascular outcomes specified in the protocol.

Secondary cardiovascular endpoints included the individual components of the primary composite endpoint, other composites of cardiovascular events (cardiovascular death, myocardial infarction, or stroke), revascularisation procedures, new-onset atrial fibrillation, new-onset diabetes mellitus, and death from any cause. Other secondary outcomes included clinical symptoms of OSA and health-related quality of life. Some of the secondary outcomes listed in the appendix (p 9) are not included in the present analysis, but will be included in future analysis.

Adverse events related to use of CPAP together with serious adverse events were assessed each time the patient was contacted during follow-up. Descriptions of the study endpoints and endpoint adjudication procedures are provided in the appendix (pp 9–12).

Statistical analysis

Assuming a 25% reduction in the likelihood of further cardiovascular events in the CPAP group compared with that in the UC group, and 10% loss of patients in follow-up, we estimated that 1264 patients (632 per group) would need to be enrolled to provide 80% power at a two-sided α level of 0·05. Additionally, for comparisons between groups, we determined that a total of 600 patients would be required in the reference group. Thus, the study required 1864 patients.

We estimated descriptive statistics of the mean (SD) with a normal distribution or median with a non-normal distribution (IQR) for quantitative variables, and we used absolute and relative frequencies for qualitative variables. We used Shapiro–Wilks to test normality of continuous variables. We compared both categorical and continuous variables in bivariate analyses with the use of corresponding tests. For the primary analysis, we used an unadjusted Cox proportional regression model according to the intention-to-treat principle based on adjudicated events. In addition, we analysed patients with multiple events by means of Poisson regression. We used one-to-one propensity score matching to match compliant patients (average of ≥4 h/night) in the CPAP group with patients in the UC group who had never used CPAP to estimate the CPAP effect in the adherence set. We analysed continuous variables (ESS, EQ-5D, and blood

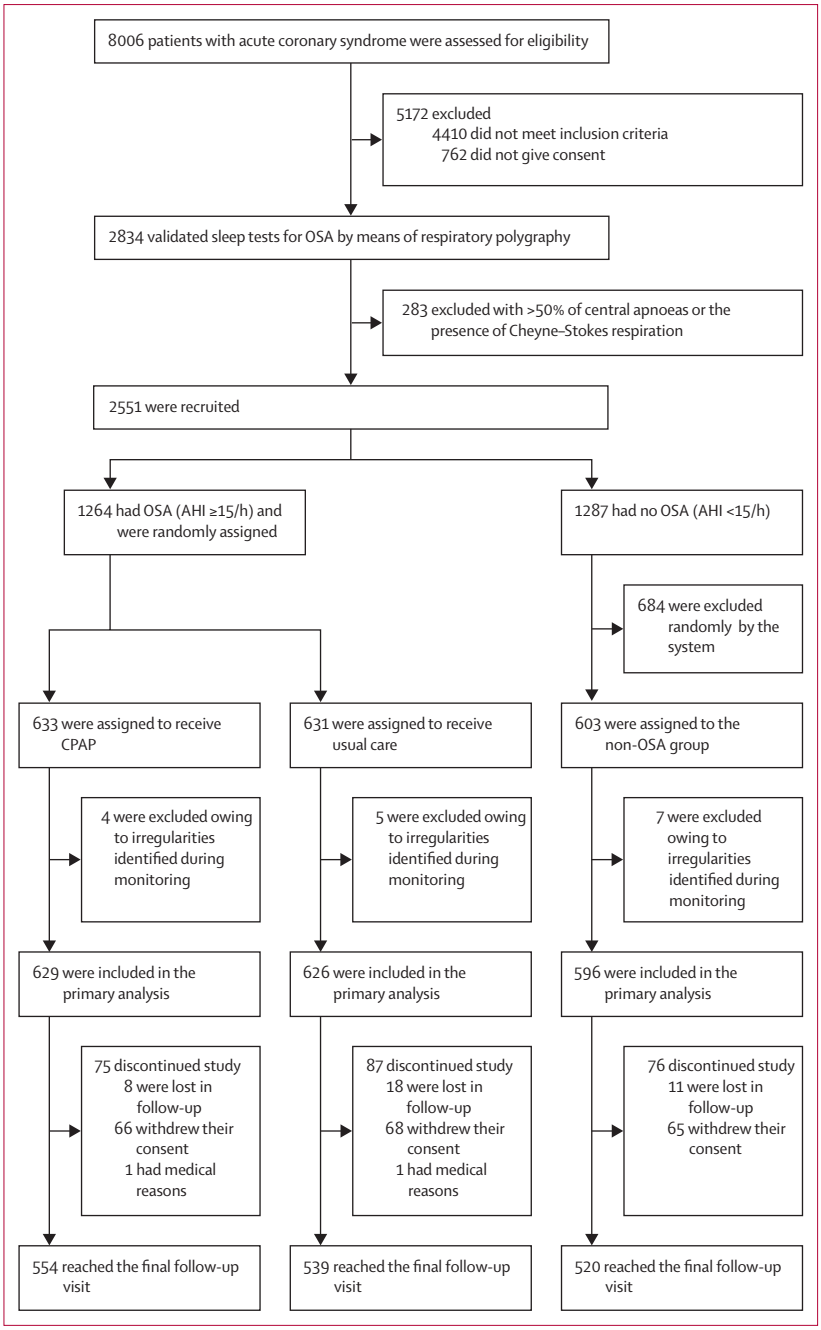


Figure 1: **Trial profile**
OSA=obstructive sleep apnoea. AHI=apnoea–hypopnoea index. CPAP=continuous positive airway pressure.

pressure) by means of mixed linear models to estimate the CPAP effect over time. Moreover, we used a Cox proportional regression model and a Poisson model, adjusted by confounder variables, to estimate the effect of OSA on the primary analysis.

We tested the proportional hazards assumption for all Cox regression models. All tests were two-sided, and we considered p values less than 0·05 as significant. Analysis was done with R statistical software, version 3.5.1 (R Project for Statistical Computing; more details in appendix pp 12–13) This study is registered as ClinicalTrials.gov, NCT01335087.

Role of the funding source

The funders of the study had no role in study design, data collection, data analysis, data interpretation, or writing of the report. The corresponding author had full access to all data in the study and had final responsibility for the decision to submit for publication.

Results

Between April 25, 2011, and Feb 2, 2018, a total of 2834 patients with ACS had respiratory polygraphy, of whom 2551 were enrolled. 1264 (49·55%) patients had an AHI of more than 15/h and 1287 (50·45%) had an AHI of less than 15/h (figure 1). Patients with ACS and OSA were randomly assigned to either the CPAP group (n=633) or the UC group (n=631). Nine patients were excluded due to irregularities identified during monitoring, leaving 1255 patients for the primary analysis (n=629 in the CPAP group; n=626 for the UC groups). Of the patients with ACS without OSA, 684 were randomly excluded and 603 were included in the reference group. 596 patients from this group were included in the primary analysis after seven patients were excluded due to irregularities identified during monitoring. The number of OSA patients who discontinued their participation in the study before the intended final visit was 162 (figure 1). The number of patients who completed 1 year of follow-up in each group was 552 (88%) in the CPAP group, 549 (88%) in the UC group, and 511 (86%) in the reference group. The median follow-up period was 3·35 years (IQR 1·50–5·31). The mean age of the patients was 60 years, and 84% were men. The mean body mass index of the patients was 29·5; the mean apneoa–hypopnoea index, 36 events per h; and the mean Epworth Sleepiness Scale score, 5·32 (table 1).

The number of patients who experienced a cardiovascular event during follow-up was not significantly different between the CPAP group and the UC group (98 patients [16%] *vs* 108 [17%]; hazard ratio [HR] 0·89 [95% CI 0·68–1·17]; p=0·40; table 2, figure 2). The annual event rate for cardiovascular events in the CPAP group was 7·18 per 100 person-years compared with 7·78 per 100 person-years in the UC group (relative risk [RR] 0·92 [0·74–1·15]; p=0·47; appendix p 20). The CPAP treatment effect was not influenced by the anthropometric or clinical characteristics of the patients. The mean CPAP adherence was 2·78 h/night (SD 2·73), and the median was 2·2 h/night (IQR 0·03–5·13; appendix pp 17, 37).

A total of 102 (17%) patients in the UC group and 90 (15%) in the reference group had a cardiovascular event during follow-up (HR 1·01 [95% CI 0·76–1·35]; p=0·93). The annual event rate of cardiovascular events was 7·62 per 100 person-years in the UC group and 6·56

per 100 person-years in the reference group (RR 1·16 [0·92–1·47]; p=0·66; appendix p 35). For more on the differences between the usual care group and reference group see the appendix (pp 15, 34–35).

We partitioned patients in the CPAP group into a poor adherence group (mean adherence <4 h/night) or a good adherence group (mean adherence ≥4 h/night) and compared these subgroups with the UC group for prevalence of cardiovascular events. 102 patients had an event (17%) of 607 patients in the UC group, 56 (15%) of 377 in the poor adherence group, and 41 (18%) of 227 in the good adherence group had a cardiovascular event (NB: 25 are missing values; for the poor adherence group *vs* the UC group, HR 0·88 [0·63–1·22]; p=0·46; for the good adherence group *vs* the UC group, HR 0·94 [0·65–1·36]; p=0·76). The results of the propensity score analysis showed a HR of 0·80 (0·52–1·23; p=0·32; appendix p 22) in the adherent subgroup (for definition of this group see appendix p 12) compared with that in the UC subgroup. Adjustment for the random hospital (site) effect did not modify these results (appendix p 23). Further details on the results for the primary endpoint in the appendix (pp 36, 38–39, 18–23).

The CPAP had no significant effect on the individual components of the primary cardiovascular composite endpoint, composite of major cardiovascular events, revascularisation procedures, all-cause death, atrial fibrillation or other arrhythmias, or newly diagnosed diabetes (table 2).

CPAP treatment was associated with a marginal improvement in ESS and blood pressure and had no effect on quality of life (figure 3). Patients in the CPAP group showed a marginally greater increase in body-mass index and waist circumference throughout follow-up compared with patients in the UC group. For more secondary and other endpoints (neck circumference, hip circumference, and serious adverse events) see the appendix (p 40).

464 (74%) of 629 patients in the CPAP group had 1538 serious adverse events and 406 (65%) of 626 patient in the UC group had 1764 serious adverse events. The most common serious adverse events were gastrointestinal disorders (113 [18%] in the CPAP group *vs* 69 [11%] in the UC group), infections and infestations (102 [16%] *vs* 88 [14%]), and musculoskeletal and connective tissue disorders (97 [15%] *vs* 104 [17%]; appendix pp 24–33).

	Continuous positive airway pressure (n=629)	Usual care (n=626)
Sex		
Female	101 (16%)	96 (15%)
Male	528 (84%)	530 (85%)
Anthropometric measures		
Age, years	59·9 (10%)	60·7 (10%)
Body-mass index, kg/m^2	29·6 (4·66)	29·4 (4·29)
Waist-to-hip ratio	0·99 (0·07)	0·99 (0·08)
Neck circumference, cm	41·0 (3·80)	41·1 (3·60)
Lifestyle risk factors		
Smoking		
Current	304 (48%)	291 (47%)
Former	172 (27%)	181 (29%)
Never	153 (24%)	154 (25%)
Drinking		
Current	174/619 (28%)	153/604 (25%)
Former	7/619 (1%)	6/604 (1%)
Never	438/619 (71%)	445/604 (74%)
Obstructive sleep apnoea measures		
Apneoa-hypopnoea index, events per h	36·4 (18·6)	35·5 (18·3)
Oxygen desaturation index >4%, per h	30·9 (24·3)	34·2 (35·5)
Mean SaO_2	91·7 (9·42)	92·0 (8·04)
Minimum SaO_2	81·9 (7·33)	81·1 (10·2)
Time with SaO_2 <90%	11·9 (18·5)	11·9 (18·8)
Epworth Sleepiness Scale	5·36 (2·50)	5·28 (2·50)
Medical history		
Hypertension	350 (56%)	357 (57%)
Diabetes	172 (27%)	157 (25%)
Dyslipidaemia	368 (59%)	350 (56%)
Stroke	17 (3%)	19 (3%)
Chronic pneumopathy	36 (6%)	38 (6%)
Neurological disease	19 (3%)	32 (5%)
Cardiomyopathy	129 (21%)	125 (20%)
First episode of acute coronary syndrome	536 (85%)	532 (85%)
Medication		
Antihypertensive drug	320 (51%)	345 (55%)
Lipid lowering drug	243 (39%)	229 (37%)
Antidiabetics oral medication	135 (22%)	126 (20%)
Insulin	40 (6%)	40 (6%)
Antiplatelet and antithrombotic drugs	170 (27%)	161 (26%)

Data are n (%), n/N (%) and mean (SD). SaO_2=oxygen saturation. Baseline characteristics for the reference group are included in the appendix (p 15).

Table 1: **Baseline characteristics**

Discussion

The ISAACC study showed that among patients with ACS and OSA, CPAP treatment did not result in a significantly lower prevalence of cardiovascular events than did UC after a median follow-up of 3·35 years. CPAP adherence did not influence these results. Additionally, the presence of OSA was not associated with a worse prognosis in patients with ACS. In contrast, treatment with CPAP was associated with marginal improvement in ESS and blood pressure.

These results should contribute to understanding the observed neutral effect of CPAP treatment for secondary cardiovascular prevention. Three previous randomised controlled trials have directly addressed cardiovascular clinical endpoints in patients who have had a major cardiovascular event. In one single-centre study, the authors assessed the CPAP effect in 244 patients with established coronary artery disease who had just had revascularisation.[22] In a second multicentre study

	Continuous positive airway pressure (n=629)	Usual care (n=626)	Hazard ratio (95% CI)	p value
Primary composite endpoint	98 (16%)	108 (17%)	0·89 (0·68–1·17)	0·40
Secondary endpoints				
Components of primary endpoint				
Cardiovascular death	12 (2%)	14 (2%)	0·83 (0·38–1·8)	0·64
Myocardial infarction	37 (6%)	35 (6%)	1·04 (0·66–1·65)	0·86
Stroke	8 (1%)	5 (<1%)	1·59 (0·52–4·85)	0·42
Hospital admission for heart failure	22 (4%)	25 (4%)	0·86 (0·49–1·53)	0·61
Hospital admission for unstable angina	34 (5%)	46 (7%)	0·73 (0·47–1·13)	0·16
Hospital admission for transient ischaemic attack	4 (<1%)	2 (<1%)	1·99 (0·37–10·89)	0·43
Other endpoints				
Composite of major cardiovascular events*	57 (9%)	50 (8%)	1·13 (0·77–1·65)	0·52
Composite for cerebral events†	12 (2%)	7 (1%)	1·71 (0·67–4·33)	0·26
Composite for cardiac events‡	91 (15%)	104 (17%)	0·86 (0·65–1·14)	0·28
Revascularisation procedures	66 (11%)	66 (11%)	0·98 (0·7–1·38)	0·91
All-cause death	27 (4%)	32 (5%)	0·82 (0·49–1·36)	0·44
Atrial fibrillation or other arrhythmias	14 (2%)	17 (3%)	0·81 (0·4–1·65)	0·56
Newly diagnosed diabetes	48 (10·71)§	39 (8·52)	1·32 (0·86–2·01)	0·20

Data are n (%), unless otherwise specified. *Cardiovascular death, myocardial infarction, or stroke. †Stroke or hospital admission for transient ischaemic attack. ‡Myocardial infarction, hospital admission for unstable angina, atrial fibrillation, or heart failure. § Based on all those who do not have diabetes at the beginning.

***Table 2*: Primary and secondary cardiovascular endpoints**

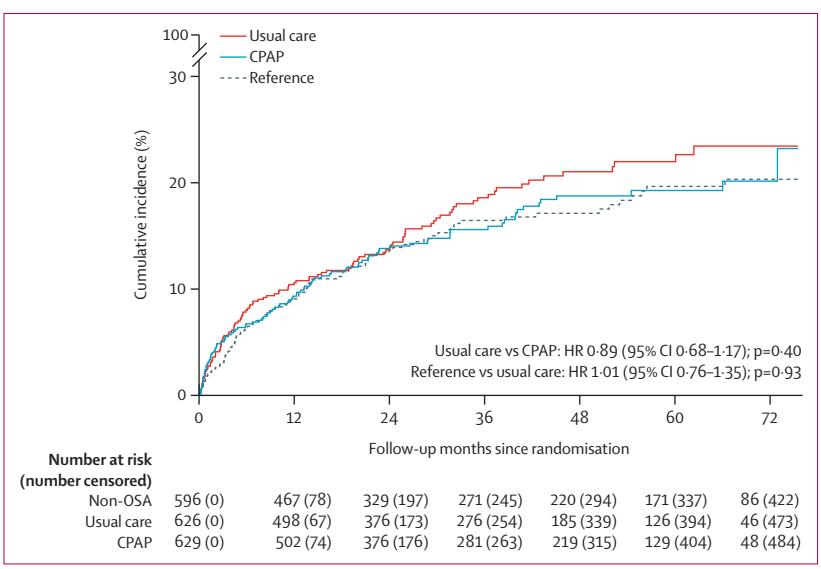


***Figure 2*: Cumulative event curve of the primary endpoint in the CPAP group, the UC group, and the reference group**
Reference=without obstructive sleep apnoea. CPAP=continuous positive airway pressure. HR=hazard ratio.

that included a total of 140 patients with ischaemic stroke, the effect of CPAP treatment on event-free survival was evaluated.[23] None of these studies showed a beneficial effect of CPAP treatment on composite cardiovascular endpoints over several years of follow-up. Finally, the SAVE study[14] is a multicentre study that recruited 2717 patients with chronic coronary artery disease or cerebrovascular disease with OSA and evaluated the effect of CPAP treatment. The primary outcome was a similar composite cardiovascular endpoint as in the ISAACC study. The results of the SAVE study showed that after a mean follow-up of 3·7 years, CPAP treatment did not reduce the prevalence of cardiovascular events. Moreover, the absence of an effect on cardiovascular outcomes was reported in multiple prespecified subgroup analyses. As previously reported, both the SAVE and ISAACC studies showed a positive effect of CPAP treatment on clinical symptoms.

A remarkable difference between the studies is that whereas the SAVE study included patients with chronic coronary artery disease or cerebrovascular disease, the ISAACC study recruited patients in the acute phase of coronary disease. Another difference is that the ISAACC study focused on a homogeneous population located in a country with high standards of health care. Notably, the SAVE study did not include a reference group of patients without OSA. The SAVE and ISAACC studies contribute to understanding the observed neutral effect of CPAP treatment for secondary cardiovascular prevention. In these two studies, CPAP treatment showed a neutral effect on patient prognosis, which was expected owing to the absence of deleterious effects of OSA on the prognosis of patients with ACS. We speculate that the potential effect of CPAP treatment would be limited only to primary cardiovascular prevention, with a greater effect observed in patients with adequate adherence than in those without adequate adherence.[14,22–24] In patients with OSA and ACS, the identification of independent predictors of recurrent cardiovascular events could

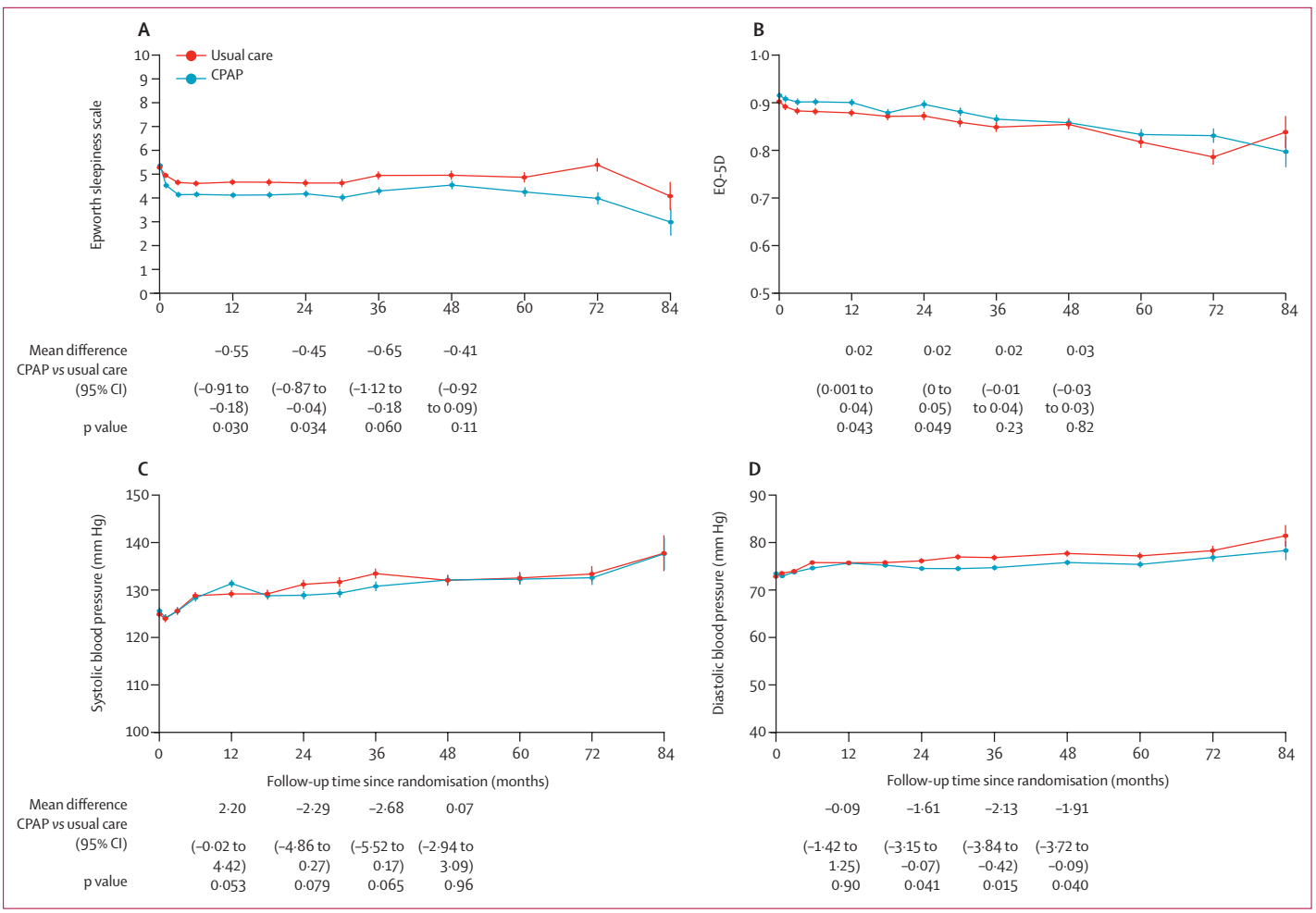


Figure 3: **Measurement of EES (A), EQ-5D (B), systolic blood pressure (C), and diastolic blood pressure (D) in the CPAP group compared with UC group throughout follow-up**
Values on the y axis are mean values for all patients at each timepoint. The mean difference between the CPAP group and UC group refers to the effect of CPAP at 12, 24, 36, and 48 months. Bars denote the SE. CPAP=continuous positive airway pressure. EES=Epworth Sleepiness Scale. EuroQol EQ-5D=an instrument for measuring quality of life.

contribute in the risk stratification for each patient, and this concept should be explored in further studies. As previously reported,[25] we observed that CPAP treatment was associated with a slight increase in body-mass index. The reasons that underline this observation have not yet been determined.

The strengths of our study include its multicentre design with a large patient population. All participating centres used the same methodology, and the sleep study was done with the same polygraph model. Nevertheless, this study has several potential limitations. First, we excluded patients with severe ACS and very poor prognosis (cardiogenic shock) as these patients have poor expectations for short-term outcomes. Additionally, we excluded patients with daytime sleepiness who exhibited the most severe OSA. From an ethical perspective, the OSA patient with an Epworth Sleep Scale score >10 should continue to be treated with CPAP. Our results apply only to non-sleepy OSA patients. Nevertheless, the number of patients excluded for these causes was relatively low (figure 1). Second, patients in this study were predominantly men; therefore, the results cannot be extrapolated to women. Third, the diagnosis is based on respiratory polygraphy, which could underestimate the severity of OSA. However, full polysomnography monitoring could be a stressful procedure for this high-risk patient group. Moreover, although full polysomnography monitoring is the recommended method for the diagnosis of OSA, data from 2017 suggest that respiratory polygraphy could also be used.[26] Fourth, patients admitted for ACS could be given pharmacological treatment for the acute phase, including treatment with opioids, that could potentially influence the results of a sleep study by increasing the AHI. Nevertheless, opioids (eg, morphine or meperidine) used for ACS have a short half-life (2 h for morphine and about 2·5–4 h for meperidine). Morphine is used almost exclusively in the first phase of the encounter to relieve chest pain. We did the sleep studies during the first 24–72 h after admission and therefore most, if not all, of the drug would have

already been metabolised. Additionally, by excluding central sleep apnoea, the potential confounding factor of opioids is avoided. Fifth, given the very low compliance to CPAP treatment (2·78 h/night), the beneficial effect of sustained well-conducted CPAP (≥4 h/night) cannot be excluded. The patients in this study did not attend any sleep unit , but were instead recruited from coronary units, without being referred for a sleep-related breathing problem or symptoms. In this population, getting a good adherence to CPAP treatment is very difficult, even for experienced research teams. However, as we have found in other studies in this area of research (the RICCADSA[22] and SAVE[14] studies), the results in the adherent and non-adherent sets do not differ. Sixth, we screened a lower number of patients than originally planned because the prevalence of OSA in the population with ACS was higher than expected and because of the exclusion of those with daytime sleepiness. And finally, we included only white patients and our results cannot be extrapolated to patients with other ethnic or racial backgrounds.

In conclusion, in a large group of patients with ACS and OSA, the use of CPAP therapy had no positive effect on the prevention of recurrent major cardiovascular events. Moreover, the presence of OSA was not associated with an increased prevalence of cardiovascular events.

Contributors

FB had full access to all of the data in the study and takes responsibility for the integrity of the data and the accuracy of the data analysis. MSdlT, ASdlT, SB, JA, JD-C, VC, OM, JMM, MLA, JFM, AB, MdlP, MM, RC, MJM, EC, SP, GR, MD, RDMcE, and FB conceived and designed the study. MSdlT, ASdlT, SB, JA, JD-C, VC, OM, JMM, MLA, JFM, AB, MdlP, MM, RC, MJM, EC, SP, GR, MD, RDMcE, and FB analysed and interpreted the data. MSdlT, ASdlT, SB, RDMcE, and FB drafted the manuscript. Critical revision of the manuscript for important intellectual content was done by MSdlT, ASdlT, SB, JA, JD-C, VC, OM, JMM, MLA, JFM, AB, MdlP, MM, RC, MJM, EC, SP, GR, MD, RDMcE, and FB. MSdlT, ASdlT, SB, and FB completed the statistical analysis. OM, LP, AC, DM, AA, and FB provided administrative, technical, or material support. MSdlT, ASdlT, and FB supervised the study.

Declaration of interests

FB received a research grant from ResMed (a Australian company that develops products related to sleep apnoea), the Health Research Fund, Spanish Ministry of Health, the Spanish Respiratory Society, the Catalonian Cardiology Society, Esteve-Teijin (Spain), Oxigen Salud (Spain), and ALLER, to develop the ISAACC trial. ResMed partly funded the study and did not have any participation or decision in study development or the present manuscript. RDMcE is funded by a Practitioner Fellowship from the National Health and Medical Research Council (Australia) and reports research funding support from Philips Respironics, Fisher, and Paykel; research equipment donations from Philips Respironics and ResMed, and speaker fees from ResMed. All other authors declare no competing interests.

Data sharing

All of the individual patient data collected during the trial will be shared. In addition, the study protocol and statistical analysis plan will be available as well. The data will be made available within 12 months of publication. All available data can be obtained by contacting the corresponding author (febarbe.lleida.ics@gencat.cat). It will be necessary to provide a detailed protocol for the proposed study, to provide the approval of an ethics committee, to supply information about the funding and resources one has to carry out the study, and to consider inviting the original authors to participate in the re-analysis.

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